

Thesis Topic Registration Form

Student's Data:

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Student's Neptun code: K1195I

Educational Information:

Training programme: programtervező informatikus, mesterképzés (MA/MSc)

I have an internal supervisor

Internal Supervisor's Name: Dr. Zsók Viktória

Supervisor's Home Institution: Department of Programming Languages and Compilers

Address of Supervisor's Home Institution: 1117, Budapest, Pázmány Péter sétány 1/C.

Supervisor's Position and Degree: Lecturer, PhD

Track within Programme: Computer Science (Software Architecture)

Thesis Title: A Comparative Testing Framework for Blockchain Layer 2 Scaling Solutions: Assessing Finality, Security, and Recovery under Adverse Conditions

Topic of the Thesis:

(Upon consulting with your supervisor, give a 150-300-word-long synopsis of your planned thesis.)

This research investigates how modern Layer 2 (L2) scaling architectures, specifically ZK Rollups, Optimistic Rollups, and Plasma/Validium chains sustain security, finality, and resilience when stressed by adverse network conditions. As Ethereum's Layer 1 (L1) evolves into a settlement and data availability layer within a rollup-centric ecosystem, understanding how L2 systems inherit, extend, or fail to uphold L1 guarantees becomes increasingly crucial.

The study addresses the question: How do different L2 mechanisms ensure finality and recover from disruptions such as L1 congestion, sequencer downtime, or malicious state proposals? It will examine key components like fraud and validity proofs, data availability, and finality propagation between L1 and L2.

The methodology involves developing a multi-L2 simulation framework that models diverse rollup architectures and reproduces challenging conditions including network partitions, delayed proofs, and mass exit events. Using Docker-based orchestration with Prometheus and Grafana for monitoring, the framework will collect comparative data across metrics such as time-to-finality, fraud detection latency, recovery cost, and data integrity under stress.

The expected contribution is both academic and practical. Academically, the research delivers one of the first systematic analyses of L2 failure modes, bridging the gap between L1 theoretical assurances and real-world reliability. Practically, it provides a decision framework for developers and enterprises to evaluate L2 technologies through measurable security-performance trade-offs. Ultimately, this work aims to strengthen blockchain scalability by clarifying how resilient multi-layer systems behave when the underlying trust foundation (the L1) is put to the test.

Budapest, 2025. 10. 14.